

Topic 8 recap

- **Nodal** and **mesh analysis** are applied to AC circuits the same way as in DC circuits.
 - Nodal equations (KCL) and mesh equations (KVL) should be written for an AC circuit in phasor domain when it is operated in steady state sinusoidal conditions.
- **Superposition** can also be applied to AC circuits with multiple independent sources in phasor domain.
 - When having sources with **different frequencies only superposition** can be used to find the steady state response of the circuit.
 - A separate phasor circuit must be solved for each frequency independently.
 - The overall response is the sum of the time domain responses of all individual phasor circuits.
 - If the **frequencies are the same**, the sum of the phasor responses will be used as the final response and then transformed back to time domain if needed.
- The concept of **source transformation** is also applicable in the frequency domain.

Topic 8 recap

- **Thevenin** and **Norton equivalent** circuits can also be used for AC circuits in phasor domain.
 - The Thevenin equivalent circuit consists of a voltage source $\mathbf{V}_{Th}$ in series with the Thevenin equivalent impedance $\mathbf{Z}_{Th}$:
 - $\mathbf{V}_{Th} = \mathbf{V}_{oc}$: Open-circuit phasor voltage across the given terminals.
 - $\mathbf{Z}_{Th} = \mathbf{Z}_{eq}$: Equivalent impedance as seen from the given terminals.
 - The Norton equivalent circuit consists of a current source $\mathbf{I}_N$ in parallel with the Norton equivalent impedance $\mathbf{Z}_N$:
 - $\mathbf{I}_N = \mathbf{I}_{sc}$: Short-circuit current through the given terminals.
 - $\mathbf{Z}_N = \mathbf{Z}_{Th} = \mathbf{Z}_{eq}$: Equivalent impedance as seen from the given terminals.
 - **Thevenin-Norton transformation** is the same as in DC:

$$\mathbf{I}_N = \frac{\mathbf{V}_{Th}}{\mathbf{Z}_{Th}} \text{ and } \mathbf{V}_{Th} = \mathbf{Z}_N \mathbf{I}_N$$

Topic 8 recap

AC Op Amps

- AC Op Amp circuits are analyzed in the same way as DC circuits.
 - Transform an AC Op Amp circuit into **phasor domain**.
 - Apply **Nodal analysis** considering **Ideal** Op Amp.
 - **Input currents** to Op Amp are **zero**.
 - **Input voltage nodes** are **equal** (with negative feedback).
- It is possible to use the formulas for common DC Op Amps in AC mode.
 - Inverting amplifier: $\frac{V_o}{V_s} = -\frac{Z_f}{Z_1}$
 - Non-inverting amplifier: $\frac{V_o}{V_s} = 1 + \frac{Z_f}{Z_1}$
 - Summing amplifier: $V_o = -\left(\frac{Z_f}{Z_1} V_1 + \frac{Z_f}{Z_2} V_2 + \frac{Z_f}{Z_3} V_3\right)$